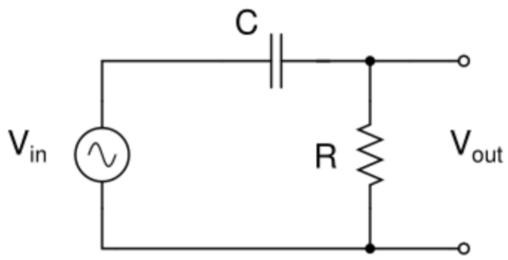
EE Dept, IIT Bombay
Academic Year: 2025-2026, Semester I (Autumn)
Course: MS101 Makerspace
EE Lecture 2: Devices & Circuits II
Sample Problems

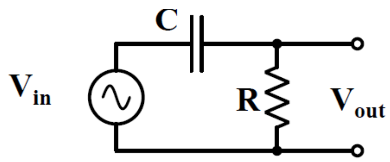
1. The circuit diagram of an RC high-pass filter is given below. For a sinusoidal input voltage, sketch the magnitude of V_{out}/V_{in} as a function of frequency. The cut-off frequency of the filter is $f_c = 1/(2\pi RC)$. Find the cut-off frequency for $C = 1\mu F$, $R = 1\ k\Omega$.



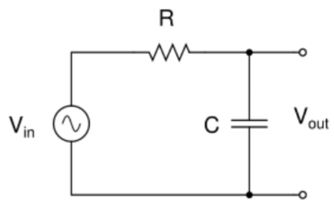
2. The circuit diagram of an RC high-pass filter is given below. Given: $R = 2\ k\Omega$, $C = 0.01\ \mu F$.

   A) What is the cut-off frequency f_c in Hz?

   B) Find the magnitude of V_{out}/V_{in} at (i) f_c and (ii) 2 kHz.



3. An RC filter circuit is shown below.



   $R = 2\ k\Omega$, $C = 0.02\ \mu F$. Its gain A is given by:

   $A = \left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{[1+(\omega CR)^2]^{1/2}} = \frac{1}{[1+(f/f_c)^2]^{1/2}}$ where $\omega = 2\pi f$ and $f_c = 1/(2\pi RC)$.

   A) Evaluate f_c in Hz.

   B) Sketch A as a function of f, indicating its values at $f << f_c$, $f = f_c$, $f_c << f$.

   C) In the above circuit, $R = 2.5\ k\Omega$ and the cut-off frequency f_c is found to be 1570 Hz. Find C in μF.

4. The current through a *pn* junction diode in the forward region can be approximated as:

   $i_d = I_S \exp [v/V_T]$,

   where v is the voltage across the diode, I_S is the reverse-saturation current, and V_T is the thermal voltage. Given: $V_T = 25$ mV, and $I_S = 10^{-13}$ A. Calculate i_d for $v = 0.6$ V. Your current should be precise to the first decimal place.

5. The current through a *pn* junction diode in the forward region can be approximated as:

   $i_d = I_S \,.\, \exp [v_d/V_T]$,

   where v is the voltage across the diode, I_S is the reverse-saturation current, and V_T is the thermal voltage. Given: $V_T = 25$ mV, and $I_S = 10^{-14}$ A.

   A resistor and this diode are connected in series, and the series combination is connected across a 5-V source such that the diode is forward biased. If the voltage across the resistor is 4.3 V calculate the diode current i_d in mA. Your current should be precise to the first decimal place.

6. The current through a *pn* junction diode in the forward direction can be approximated as

   $i_d = I_S \,.\, \exp [v_d/V_T]$,

   where v is the voltage across the diode, I_S is the reverse-saturation current, and V_T is the thermal voltage. Given: $V_T = 25$ mV, and $I_S = 10^{-12}$ A.

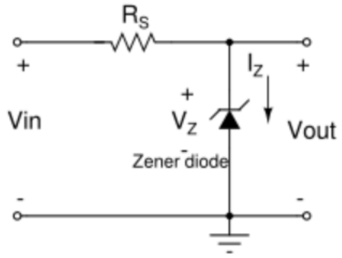
   This diode is connected in series to a resistance of 1 kΩ, and the series combination is connected across a 5-V source such that the diode is forward biased.

   A) Assume a diode forward voltage drop of $v_d = 0.6$ V. Calculate the current i_d in mA.

   B) For the above problem, obtain better estimates of the voltage v_d and current i_d through numerical iterations. [Hint: From the expression $i_d = I_S \,.\, \exp [v_d/V_T]$, we obtain $v_d = V_T \ln (i_d/I_S)$. Start with $v_d = 0$ and obtain a starting value for i_d . Obtain v_d for this i_d value. Find i_d corresponding to this v_d, and so on. Continue this iterative process about three times to obtain fairly steady values for v_d and current i_d.]

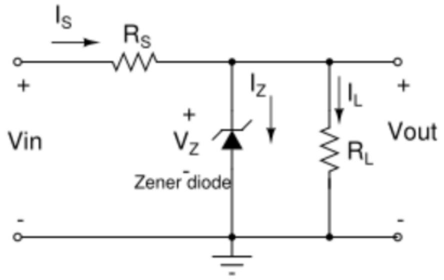
7. A Zener diode circuit is shown below. The Zener voltage $V_z = 6$ V, and the Zener I-V characteristic is linear in the Zener region with a slope of 200 Ω. Given: $V_{in} = 12$ V, $R_S = 800$ Ω.

   Evaluate I_Z in mA and V_{out} in volts. [Hint: Equivalent resistance of the Zener diode in the Zener region = 200 Ω. Replace the Zener diode with a voltage source equal to V_Z in series with the Zener equivalent resistance. Now evaluate I_Z and then obtain V_{out}.]



8. A Zener regulator circuit is shown below. Given: V_{in} = 20 V, R_S = 500 Ω, R_L = 2 kΩ, Zener voltage V_Z = 12 V, and equivalent resistance of the Zener in the Zener region = 200 Ω.

Evaluate the Zener regulator output V_{out} in volts and the currents I_S, I_Z, and I_L in mA. [Hint: Replace the Zener diode with a voltage source equal to V_Z in series with the Zener equivalent resistance. Apply superposition method to obtain V_{out} and then obtain the currents.]



9. In the Zener regulator circuit shown in Q8, evaluate the following:

A) V_{out} when $R_L = \infty$ (i.e. R_L is removed from the circuit).

B) V_{out} when R_L= 100 Ω. [Hint: When R_L = 100 Ω, the Zener diode will not operate in breakdown, since [$V_{in} \cdot R_L/(R_L + R_S)$] < V_Z. Hence $V_{out} = V_{in}\, R_L/(R_L + R_S)$.]

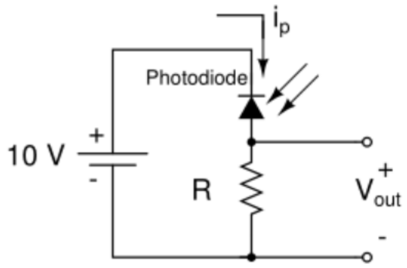
C) Based on the observation in part (B) above, obtain the smallest value of R_L that can be employed in the Zener regulator circuit of Q8.

10. Assume that a red LED having a forward voltage drop of 1.8 V is used as a power indicator by connecting it in series with a resistance to a Li-ion battery of 3.7 V. Choose the most suitable resistance value out of the following standard resistances given to you: 100, 120, 150, 180, 220, 270, 330 Ω such that the current through the LED is close to 10 mA.

11. A photodiode circuit is shown below where it is reverse-biased with a 10 V battery and R = 1 MΩ. The photodiode has a dark current of 100 nA. The photodiode has a responsivity of 0.5 A/W at 630 nm, the wavelength corresponding to red light.

[Note: Dark current is the photocurrent i_p when no light is falling on the photodiode (same as the reverse saturation current). Responsivity is the photodiode's gain parameter, which is ratio of the photocurrent to the optical power of the incident light.]

a) Evaluate V_{out} under no-light conditions.
b) Evaluate V_{out} when 4 μW of optical power falls on the photodiode.



12. Mark all the correct options.

A) Gallium Arsenide (GaAs) is the most commonly used semiconducting material for making rectifier diodes.

B) Silicon (Si) is the most commonly used semiconducting material for making rectifier diodes.

C) Germanium (Ge) is the most commonly used semiconducting material for making rectifier diodes.

D) The reverse saturation current in a typical rectifier diode is of the order of 1 mA.

E) The reverse saturation current in a typical rectifier diode is of the order of 1 μA.

F) The reverse saturation current in a typical rectifier diode is of the order of 1 nA.

G) The reverse saturation current in a typical rectifier diode is of the order of 1 pA.

13. Mark all the correct options.

A) Zener diodes are commonly used for rectifier applications in DC power supplies.

B) Zener diodes are commonly used as voltage references in DC power supplies.

C) Zener diodes do not conduct when they are forward-biased.

D) Zener diodes behave like ordinary diodes when they are forward-biased.

E) A Zener diode having a steep I-V characteristic in the Zener region would imply very small Zener resistance.

F) A Zener diode having a steep I-V characteristic in the Zener region would imply very large Zener resistance.

G) Zener diodes with smaller Zener resistances in the Zener region are better for voltage regulation applications compared to those with larger resistances.

H) Zener diodes with larger Zener resistances in the Zener region are better for voltage regulation applications compared to those with smaller resistances.

14. Mark all the correct options.

A) LEDs are special diodes with transparent packaging that emit light when reverse-biased.

B) LEDs are special diodes with transparent packaging that emit light when forward-biased.

C) The light in an LED is produced through non-radiative recombinations.

D) The light in an LED is produced through radiative recombinations.

E) LEDs are made using semiconductor of the types known as direct bandgap materials, such as Gallium arsenide (GaAs).

F) LEDs are made using semiconductor of the types known as indirect bandgap materials, such as Gallium arsenide (GaAs).

G) Wavelength of the light emitted by an LED depends on the current passing through it.

H) Wavelength of the light emitted by an LED depends on the electronic bandgap of the material used.

15. Mark all the correct options.

A) Photodiodes are generally forward-biased for use as light sensors.

B) Photodiodes are generally reverse-biased for use as light sensors.

C) Photocurrent in a photodiode is directly proportional to the intensity of the incident light.

D) Photocurrent in a photodiode is inversely proportional to the intensity of the incident light.

E) Photocurrent in a photodiode is constant irrespective of the intensity of the incident light.

16. Mark all the correct options.

A) Solar cells are illuminated photodiodes without reverse bias.

B) Solar cells are illuminated photodiodes with reverse bias.

C) Solar cells convert incident light energy into electrical energy based on the photoelectric effect.

D) Solar cells convert incident light energy into electrical energy based on the photovoltaic effect.

E) For proper operation, solar cells must be reverse-biased using a battery or any other DC voltage source.

F) For proper operation, solar cells do not require any battery or any other DC voltage source.

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